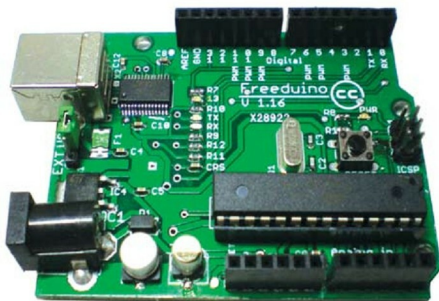


ARDUINO AND FREEDUINO OPEN AUTOMATION

Arduino and Freeduino microcontroller boards are designed using the concept of open source. Get started with them for developing your own automation applications



Freeduino board

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All of you must have read about this fantastic concept called 'open source software.' Here I share with you the concept of open source hardware.

Open source is a development methodology in which a product is developed free of cost, designed free of cost and distributed free of cost. That means it gives practical accessibility to a product's source (goods and knowledge).

Open source hardware refers to computer and electronic hardware that are designed in the same fashion as Free and Open Source Software (FOSS). Arduino and Freeduino microcontroller boards are two examples of the open source hardware.

What do Arduino and Freeduino do?

The Arduino microcontroller board is designed using the concept of open source. It is based on very simple, flexible hardware and software. It can receive multiple inputs and you may control multiple devices like lights and motors. In a nutshell, using Arduino you may automate almost anything.

The microcontroller chip fitted on the board can be easily programmed using a high-level language like C or C++. Many people have started using Ruby-on-rails for Arduino. Arduino stands out against the rest of the boards as it can easily communicate with computer software like Flash, Processing and MaxMSP. These boards can be easily built by

hand or purchased preassembled. The software IDE (integrated development environment) can be downloaded for free from Arduino website.

The Freeduino microcontroller board is a clone of Arduino. It is fully compatible with Arduino hardware and runs the same software.

Arduino and Freeduino differ in that Arduino is a protected trademark and Freeduino is trademark-free. This means you can make new products, brand them as Freeduino and sell them—or do whatever you want with them—without asking.

What else can you do?

Well, sky is the limit; it's time to put on your creative hats. So far I have seen, heard and found on the Internet that people have done lots of amazing things with these little giants.

Arduino/Freeduino have already been used for:

1. Robotics and motor (servo, stepper, solenoid) control
2. Real-time audio/video processing (cameras, MIDI, synthesizers, polyphony)
3. Infrared-/radio-controlled toys and gadgets
4. Interfacing to mice, keyboards, joysticks/game pads and even iPhones and SD/MMC/Compact Flash memory cards
5. Tagging and labeling with radio frequency identification and barcode
6. Sensor connectivity based on distance, pressure, temperature, compass, GPS (global positioning system) and motion
7. Communication protocols like